

Portfolio *of projects*

Development of chatbots, web services and Alice skill builders

Projects delivered by the Webguru.pro team in 2017–2020

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CTO at Webguru.pro · Senior PHP / Go Developer

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Webguru.pro and my role

Turnkey development of chatbots, Alice skills and web services

Webguru.pro was a development team active in 2017–2020, specialising in chatbots for Telegram / Viber / VK / Facebook Messenger, Yandex.Alice skills, web services and turnkey CRM systems. Clients ranged from local businesses to federal-scale companies.

MY ROLE

CTO at Webguru.pro

Chief Technical Officer: service architecture design, tech-stack decisions, leading the development team, code review, and hands-on implementation of core modules (middleware, billing, dialogue builder).

30+

delivered
projects

15+

Telegram
bots

12

repositories
analysed

3

letters of
appreciation

FLAGSHIP PROJECT · 01

AliceBot.pro

A SaaS skill builder for the Alice voice assistant (Yandex) with a multi-platform core — Alice + Telegram + Viber + Facebook Messenger

20+

models
in Phalcon ORM

20+

controllers
for the backend

4

channels
on one platform

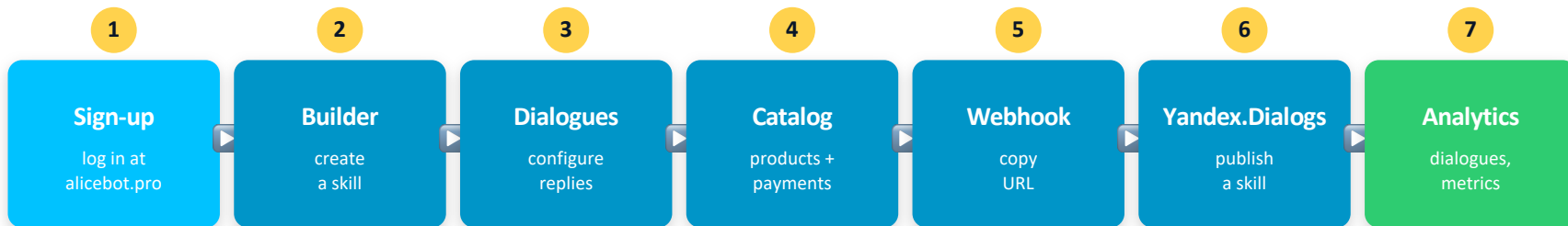
SaaS

tariffs, token
sessions, Robokassa

→ alicebot.pro · Press coverage: spark.ru/startup/webguru-pro/blog/45356

How a client builds an Alice skill

A 7-step process in a visual builder, with no coding required



WHAT IT SOLVES FOR BUSINESS

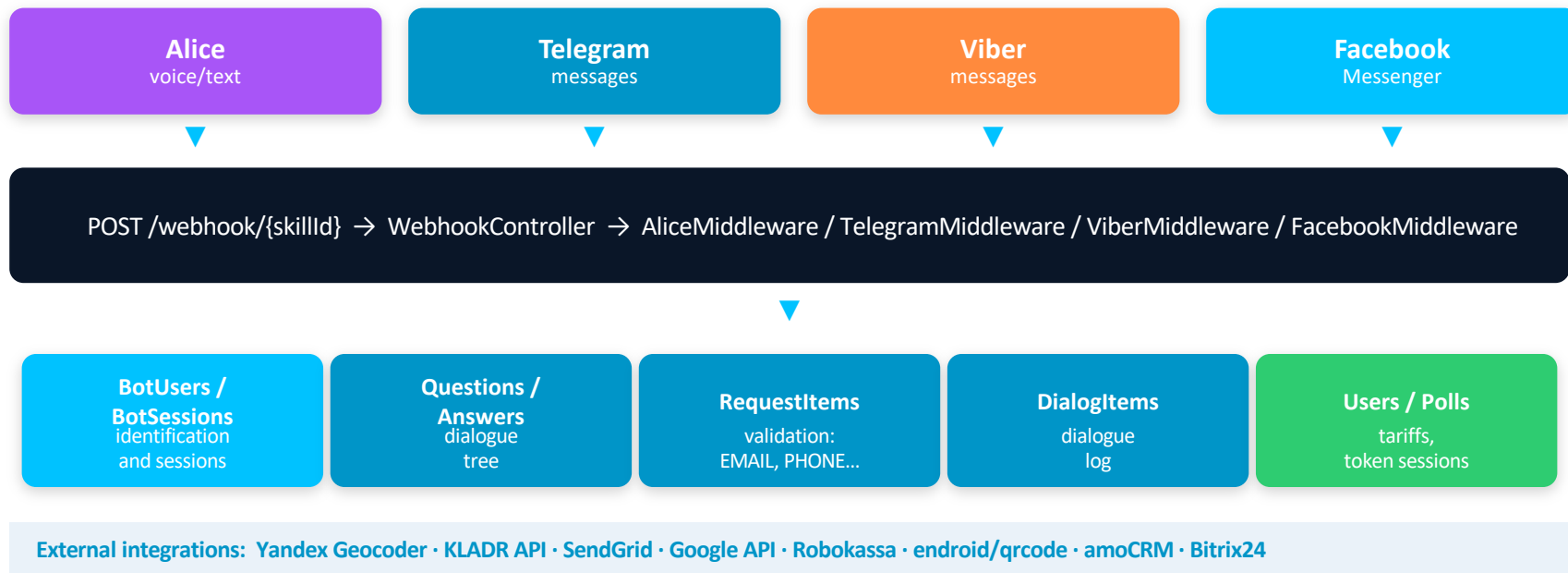
- call-center offload — routine questions handled by Alice
- conversion uplift — interactive dialogue instead of banners
- customer surveys and analytics
- sales through a voice interface

WHAT THE CLIENT GETS

- visual editor for the dialogue tree
- request analytics and dialogue history
- product catalog and payment processing
- subscription-based pricing
- integration with amoCRM and Bitrix24

Webhook-request processing flow

Phalcon MVC · 4 channels · a single middleware core



Incoming-message processing algorithm

AliceMiddleware::call — detailed flow taken from the source code



Dialogue-builder algorithm

QuestionAnswerTreeBuilder — question-answer tree with conditional transitions

HOW THE DIALOGUE ENGINE WORKS

1

Questions — tree node

question text, button type, pointers to next questions per answer branch (conditional transitions)

2

Answers — reply options

button / text / inline gallery · linked to Questions via next_question_id

3

10 input-validation types

EQUAL · NOT_EQUAL · EXISTS · NOT_EXISTS · NOT_EMPTY · DATE · TIME · PHONE · EMAIL · NUMBER

4

Text placeholders

templates with placeholders {name}, {phone}... auto-filled from RequestItems

EXAMPLE: DIALOGUE TREE

Hi! Your name?

▼ validation (text)

Phone?

Error — re-ask

▼ PHONE validation

Order placed

Validation fails → fall back to the same node and re-ask. Success → follow next_question_id. History is logged to DialogItems for analytics.

Platform capabilities and tech stack



Visual dialogue builder

question-answer tree, 10 validation types, conditional transitions



Multi-platform core

Alice / Telegram / Viber / Facebook / VK Middleware — single backend



User sessions

BotUsers + BotSessions, dialogue-context restoration, TTL 1800 sec



SaaS tariffs + Robokassa

tariff field in Users, RoboController with MD5 signing, success/fail endpoints



Geocoding and addresses

YandexGeoService · KladrService — address auto-complete via KLADR API



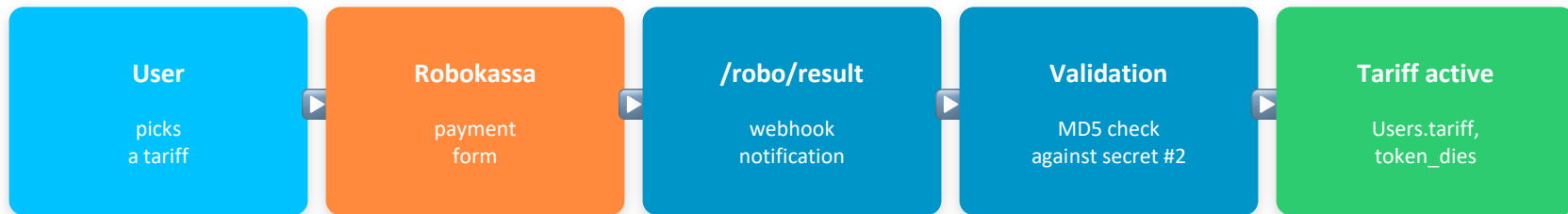
Email campaigns and QR codes

MailService + EmailsService (SendGrid) · endroid/qrcode for offline use

STACK: PHP 7 · Phalcon Framework 3 · MySQL · Phalcon\Mvc\Model · yii2-migrate · telegram-bot/api · viber-bot-php · fb-messenger-php · endroid/qrcode · SendGrid · Google API Client · Robokassa

Robokassa integration: paying for a tariff

RoboController → MD5 signature → notification validation → activation



SIGNATURE VERIFICATION ALGORITHM

- Robokassa sends a POST with parameters OutSum, Invid, Shp_item, SignatureValue
- the controller computes the expected signature: MD5(OutSum:Invid:password2:Shp_item=...)
- compares to the incoming SignatureValue in upper case; mismatch → "bad sign", exit
- on match — marks the invoice paid, activates the tariff (Users.tariff) and refreshes token_dies
- returns "OK{Invid}" — the format Robokassa accepts as confirmation

HR survey bot for retail staff

For retail chains: automated first-stage candidate screening

Automated first-stage screening. The chatbot walks a candidate through the application form, validates each field, and e-mails the candidate card to the store branch.

Usage scenario: a QR code with a unique link to the vacancy is placed at the store entrance; the candidate scans it and completes the form inside Telegram. This replaces the first HR phone call and works 24/7.

Result: irrelevant candidates are filtered out before a real HR manager is involved, saving 50–70% of the recruiting team's time.

QR

unique code
per vacancy

4

channels:
TG, Viber, VK, FB

10

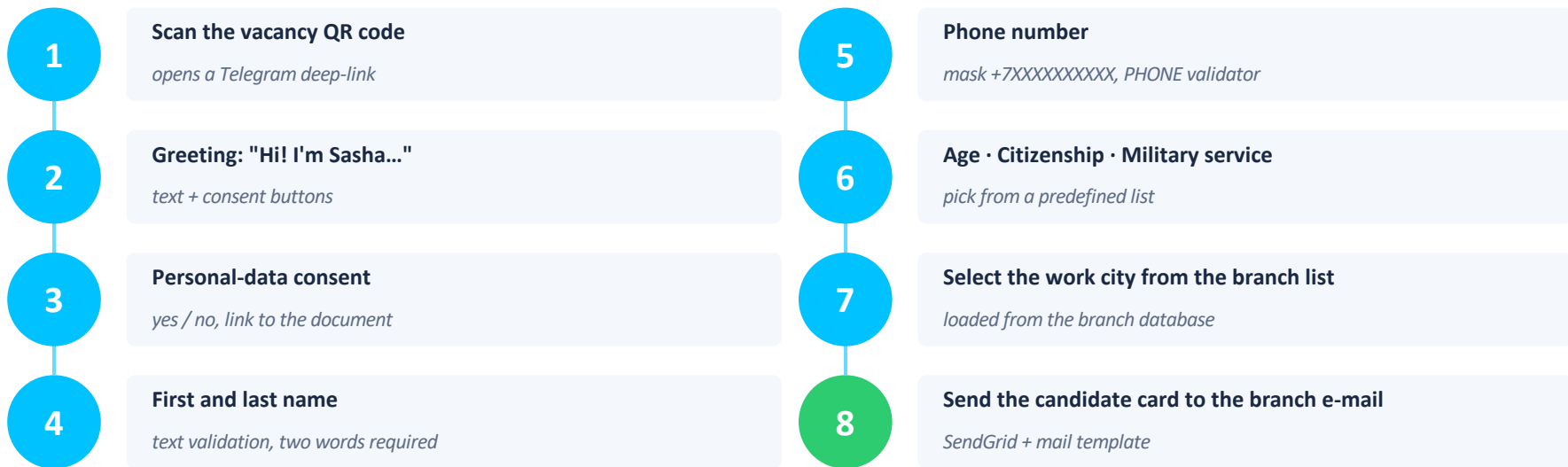
validation types
for form fields

auto

sends the card
to the branch e-mail

Candidate application-form algorithm

Sequence from the bot specification · 8 steps · validation at every stage



STACK: PHP 7 · Phalcon · MySQL · Telegram/Viber/FB Bot API · endroid/qrcode · KLADR · SendGrid

Admin-panel functionality

For HR managers — a survey builder and vacancy editor



Survey builder

question tree, answer types (text, button, geo, photo), validation rules and transitions between questions



Vacancy editor

title, description, salary, region, city, link to a questionnaire



QR-code generation

android/qrcode — for offline stands: printable QR with a deep-link to the vacancy in the bot



Branch addresses

E-mail templates with dynamic variables, recipients linked to a specific vacancy



Response reporting

vacancy funnel: how many started → how many completed, drop-off per step

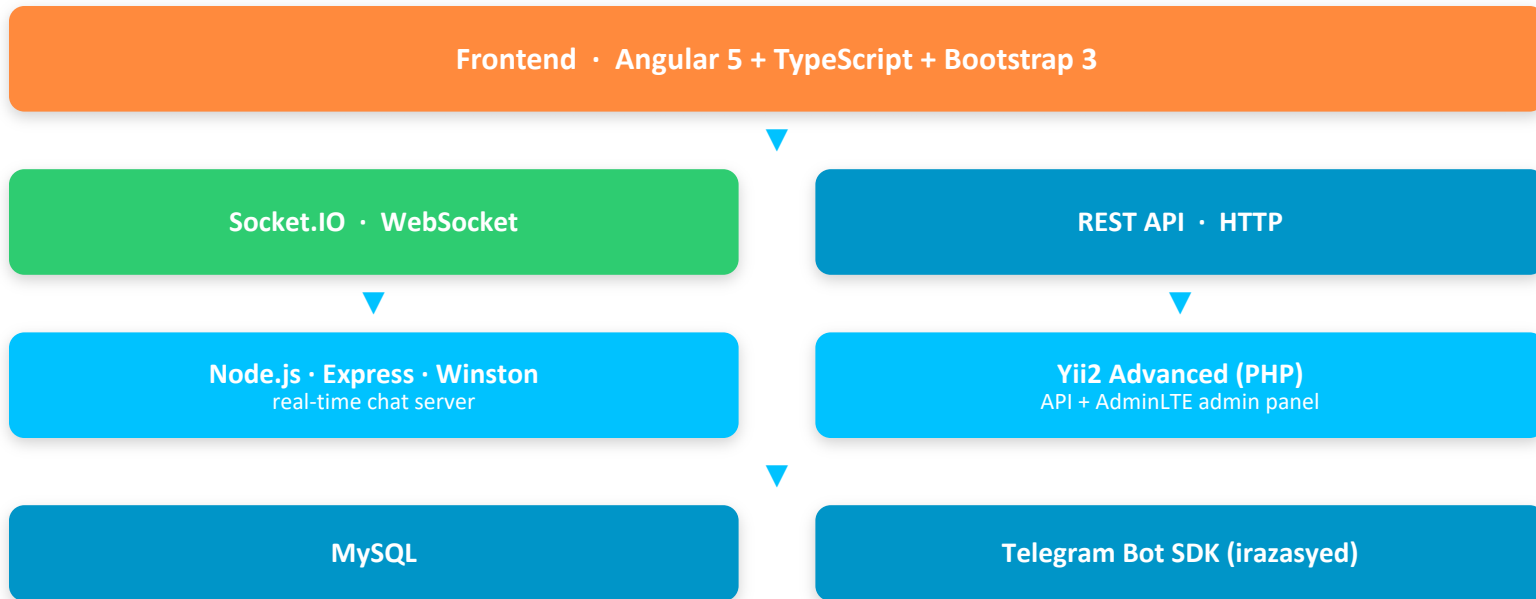


Candidate cards

view every candidate answer, tagging, Excel export for the HR team

Live chat hotline for Alliance Retail Security

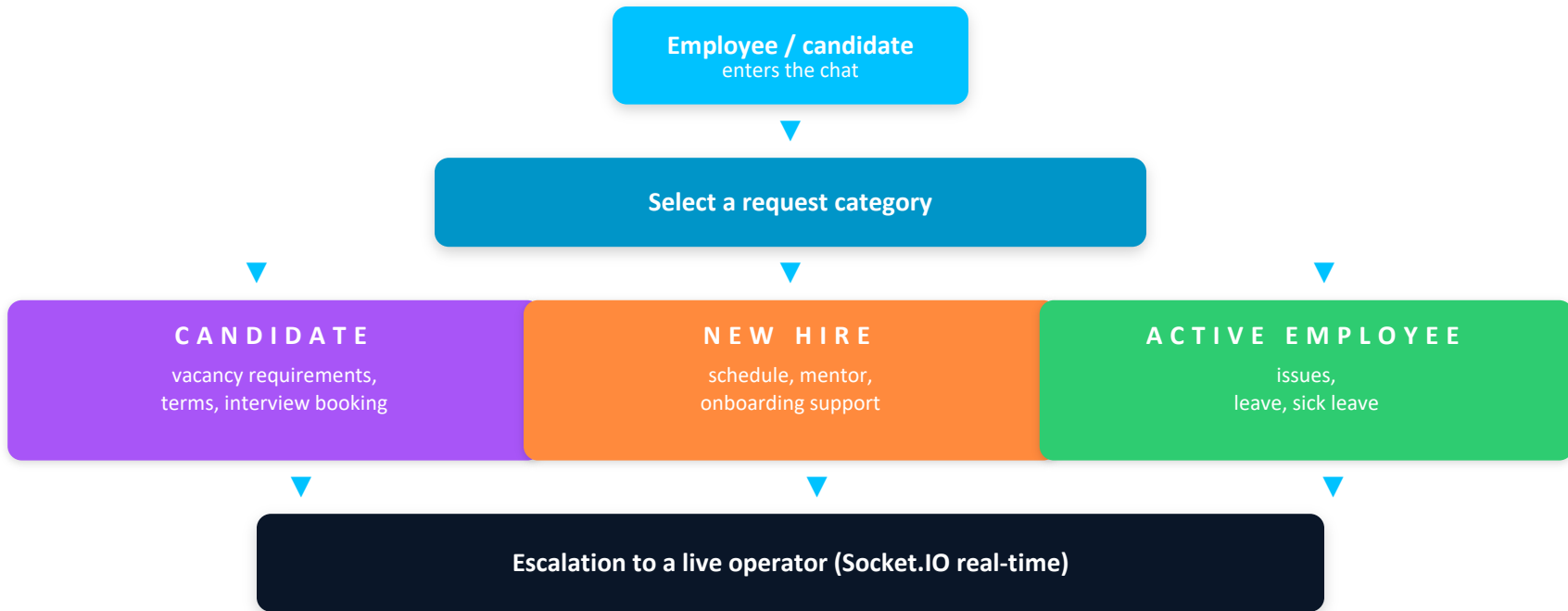
Real-time chat for a retail-chain security service · 3-tier architecture



Task: a 24/7 hotline for front-line security staff — quick answers to typical questions + escalation to a live operator

Employee hotline interaction algorithm

3 parallel scenarios: candidate / new hire / active employee



vl-taxi.ru — taxi booking across Crimea

Website + CMS + Telegram notifications · Phalcon Framework 3.4



SEO HIGHLIGHTS

- transliterated city URLs: vl-taxi.ru/bahchisaray
- route URLs: /taksi-bahchisaray-yalta
- meta-tag templates with #city1, #city2, #price
- subdomains for priority cities
- sitemap.xml generation

STACK (FROM SOURCE CODE)

- PHP 7, Phalcon Framework 3.4
- MySQL + Phalcon\Mvc\Model (ActiveRecord)
- 5 models: Cities, Destinations, Tariffs, Reviews, Phones
- SendGrid (e-mail notifications)
- Codeception (acceptance + functional tests)

Content management system — 12 admin modules

Modules for editing content, rates, SEO and orders



Cities

city CRUD, transliterated URLs



Routes

city pairs, prices, flags



Rates

Eco/Com/Com+/Uni/Biz/Mini



Site settings

overall configuration



Reviews

moderation and publishing



Promo codes

discounts and campaigns



Add-ons

child seat, booster



Orders

orders and their statuses



SEO settings

meta-tag templates



Callback

call-back requests



Help

help articles and FAQ



Taxi phones

dispatcher contacts

Karpala.ru — taxi-order marketplace

Web service + Telegram notifications + Android app · Yii2 Advanced

Dispatcher
creates an order

Driver
looks for orders

Admin
manages the marketplace

ORDER MARKETPLACE · Yii2 Advanced · MySQL · REST API

intake · storage · filtering · calculation · order logging

Website
karpala.ru

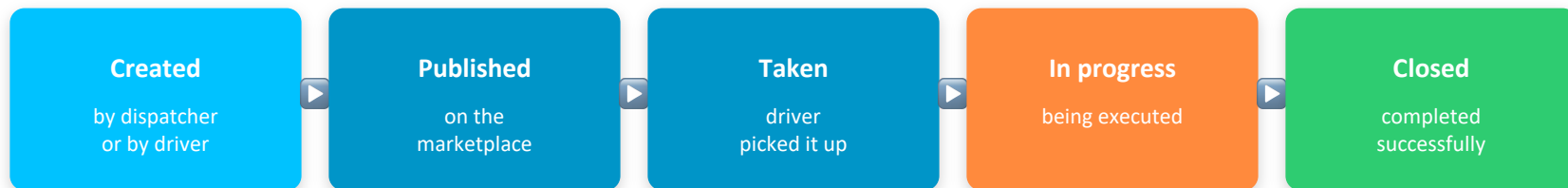
Telegram notifications
new orders for drivers

Android app
Google Play

STACK: Yii2 2.0.13 · Telegram/Viber/FB Bot API · Guzzle 6 · Gregwar\Image · Codeception · MySQL · Android (Java)

Order-lifecycle algorithm

State machine from creation to closing · Telegram integration



WEB SERVICE

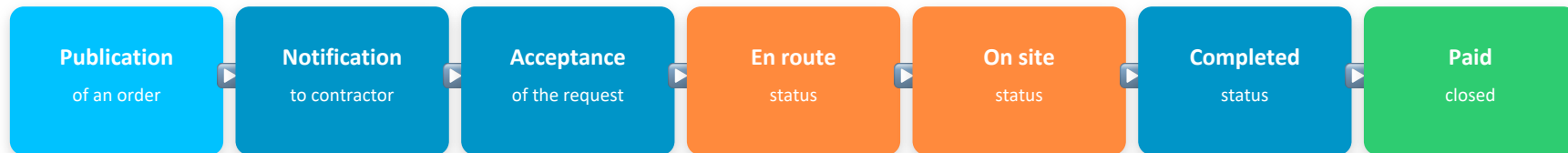
- dispatchers create orders
- drivers publish orders
- drivers purchase orders
- filtering by date, rate, route
- reporting
- system administration

ANDROID APP

- Google Play listing
- filtered order list
- push notifications via Telegram
- one-tap order acceptance
- routing via external maps
- REST sync with the web service

StartBiz.space — HRM platform for freight services

Client ↔ contractor marketplace via a Telegram bot · Fat-Free Framework



BOT SIDE

- client publishes a task
- Telegram broadcast of new offers
- client approves the contractor
- state machine with 7 states

ADMIN SIDE

- contractor rating and black list
- moderated contractor sign-up
- admin panel: contacts, orders, stats
- rules and news for subscribers

STACK: PHP · Fat-Free Framework (F3) · MySQL · telegram-bot/api · ContactApi · socket server

Three user roles, their functions and relationships

Client — Contractor — Admin

Client

- publishes a new order
- picks from responding contractors
- tracks status in real time
- rates the contractor and closes the deal

Contractor

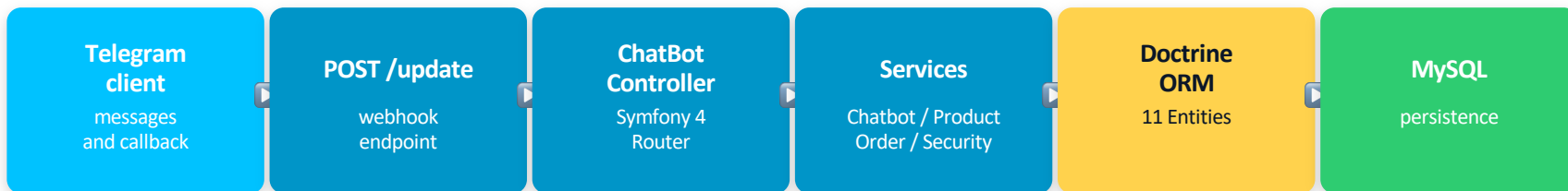
- sign-up and moderation
- receives order notifications
- accepts or declines a request
- updates status via the Telegram bot

Admin

- moderates new contractors
- manages the black list
- views order stats
- publishes news and rules

BraidsBot — Telegram e-commerce store

Full e-commerce experience inside the messenger · Symfony 4 + Doctrine ORM



11 DOCTRINE ENTITIES



STACK: Symfony 4 · Doctrine ORM · EasyAdmin · Twig · SwiftMailer · telegram-bot/api · Monolog · MySQL

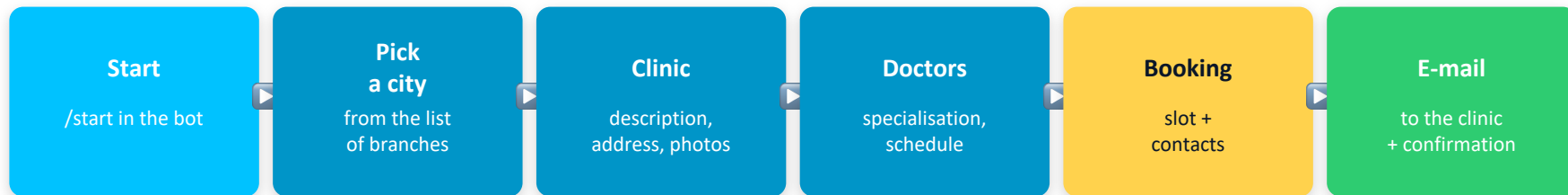
Order-placement algorithm in Telegram

From /start to order completion via inline keyboards · 8 steps



Centereko_Bot — bot for the "Centr EKO" clinic network

Patient appointment booking across Russia · Symfony 4 + EasyAdmin



ENTITIES (DOCTRINE)

- Clinic — a clinic (address, photos, opening hours)
- Doctor — a physician (name, specialisation, schedule)
- Chat — dialogue context with the patient
- Post — clinic articles / news
- RecipientEmail — notification addresses
- User — EasyAdmin admin users

STACK

- Symfony 4 + Doctrine ORM + EasyAdmin
- Vich Uploader — clinic photo uploads
- Pagerfanta — pagination
- SwiftMailer — patient notifications
- telegram-bot/api
- phpstan + php-cs-fixer (code quality)

Features for patient and clinic

Two user scenarios: patient ↔ clinic administrator

PATIENT

- pick a city from the branch network
- browse clinics with photos
- list of doctors with specialisations
- view a doctor's schedule
- get directions to the clinic
- book an appointment inside the bot
- appointment confirmation by e-mail

CLINIC ADMINISTRATOR

- clinic CRUD via EasyAdmin
- photo uploads (Vich Uploader)
- manage the doctor list
- edit schedules
- publish articles and news
- manage notification recipients
- view patient booking history

CRM Raskleika — a micro-CRM for field services

Tracking ad-pasting fieldwork with photo reports



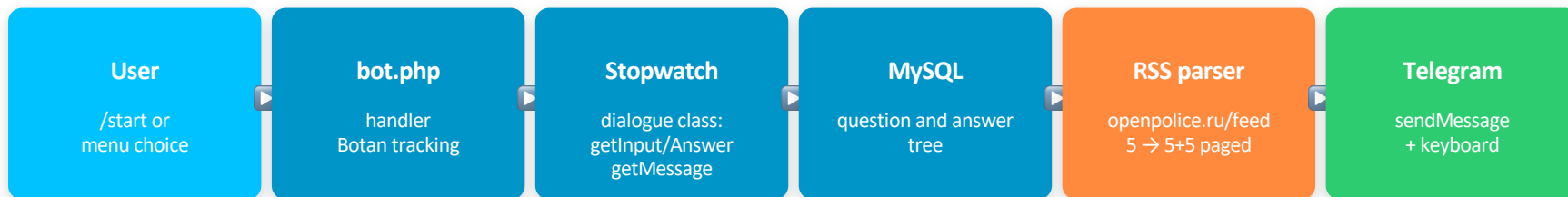
PROJECT SOLUTION

Problem: contractors upload 50–100 photos per order, but the default PHP upload is capped at 20 files. Solution — AJAX uploads one file at a time with progress, plus a separate Java module that handles large batches over a native protocol. The system archives photos and links them to orders in the database.

STACK: PHP + Fat-Free Framework · MySQL · AJAX upload · SMTP (email) · separate Java module

Openpolice_ruBot — legal-aid Telegram bot

Interactive quiz for the openpolice.ru portal · native PHP + RSS



BOT CONTENTS

- an interactive quiz on the correct behaviour during police contact
- case studies: an officer approaches you · your statement is refused · someone knocks at your door
- legal guidance for every scenario
- "Project News" section — RSS feed from openpolice.ru paginated in chunks of 5

STACK: Native PHP 7 · telegram-bot/api 2.2.4 · Botan (analytics) · MySQL · SimpleXML (RSS) · cURL

Corporate Telegram bots

AB InBev · President-Group · Braslet-Service — 3 projects with different goals

Abinbev_official_bot

AB INBEV · HR · 2 COUNTRIES

The official chatbot of AB InBev in Russia and Ukraine.

- career opportunities and internships
- divisions and brand descriptions
- RU/UA localisation
- "help the bot improve" feedback

PresidentLegal_Bot

PRESIDENT-GROUP · LEGAL

A bot for the President-Group law firm — consultations and bookings.

- company and service info
- book a lawyer appointment
- FAQ on common questions
- service catalog and consultation

Brasletservice_Bot

BRASLET-SERVICE · E-COMMERCE

A bot for Braslet-Service — order goods and interact with customers.

- bracelet catalog
- order and delivery flow
- route to the office
- contact a manager

C A T A L O G · 1 2

Telegram bots

8 highlights from the catalog

Concierge, travel, event tickets, pharma, municipal projects, HR promo and a driver marketplace.

Stack per project: PHP · telegram-bot/api · industry-specific APIs per bot

Telegram bots — project catalog

Description · functionality · project closed



MyConciergeBot

CONCIERGE · TRAVEL

Traveller's concierge: guide, taxi booking, translator.

- multi-language support
- what's-on and city guide
- translator for 6 languages, weather, exchange rates



TravelRuBot

TRAVEL · TOURS

Hot-tour picker integrated with level.travel.

- tour search and booking
- weather forecast
- currency converter



TicketsRuBot

TICKETS · EVENTS

Event-ticket search — integrated with ponominalu.ru.

- concerts · exhibitions · museums
- booking and purchase
- event cards



Clean Podmoskovye

CIVIC · MUNICIPAL

A municipal service for reporting illegal waste dumps.

- photo/video upload of the violation
- content moderation
- phone-balance top-up · Top-5

Telegram bots — project catalog

Description · functionality · project closed



KagocelBot

PHARMA · PROMO

A pharma promo bot: find nearby pharmacies via geolocation.

- nearest-pharmacy search (geo)
- product quiz
- geolocation handling



Gamethrones_bot

EMPLOYER · PROMO

A HeadHunter promo bot built by the Braininglab agency.

- user surveys
- stats collection and display
- custom UI theme



drivers73Bot

HR · DRIVERS

A driver marketplace bot inside startbiz.space.

- receive new orders
- statuses: en route / on site / done
- payment confirmation



BotSupport

SUPPORT · CROSS-PROJECT

A service channel that notifies managers across all Webguru.pro projects.

- bot-failure alerts
- new-order notifications
- broadcasting updates to clients

Consolidated tech stack across projects

Based on source-code analysis · 12 repositories

BACKEND FRAMEWORKS

- Phalcon Framework 3.4 — AliceBot · HR · vl-taxi
- Symfony 4 + Doctrine — BraidsBot · Centereko
- Yii2 Advanced — karpala · ARS
- Fat-Free (F3) — CRM · StartBiz · AB InBev
- Native PHP 7 — Openpolice

BOT APIs

- telegram-bot/api (all Telegram bots)
- irazasyed/telegram-bot-sdk (ARS)
- bogdaan/viber-bot-php
- pimax/fb-messenger-php
- Yandex.Dialogs API (AliceBot)

DATABASES & ORM

- MySQL — all projects
- Doctrine ORM — Symfony
- Phalcon\Mvc\Model — AliceBot · vl-taxi
- Yii2 ActiveRecord — karpala · ARS
- Migrations: yii2-migrate

REAL-TIME & FRONTEND

- Socket.IO + Node.js + Express + Winston (ARS)
- Angular 5 + TypeScript + Bootstrap (ARS)
- Twig — Symfony templates
- Gulp + SCSS (prototyping)
- jQuery + Chart.js + jqPlot (admin panels)

INTEGRATIONS

- Yandex Geocoder + KLADR — AliceBot · HR
- endroid/qrcode — QR-code generation
- SendGrid — e-mail campaigns
- Robokassa — payments (AliceBot)
- amoCRM · Bitrix24 · Google Sheets

MOBILE & TOOLING

- Android app (karpala · Java · Google Play)
- Codeception — tests (vl-taxi · karpala · ARS)
- phpstan + php-cs-fixer (Centereko)
- PHPUnit (BraidsBot)
- Gulp + Babel + BrowserSync (front prototypes)

Applied architectural patterns

Recurring design decisions across several projects



Middleware chain

base BotMiddleware + 5 platform-specific subclasses (AliceBot)



MVC + ActiveRecord

Phalcon · Symfony · Yii2 — controllers
→ services → ORM models



Finite State Machine

order and dialogue states: karpala,
startbiz, HR bot, BraidsBot



Webhook + signature

MD5 validation for Robokassa · token-authenticated Telegram webhooks



WebSocket + REST

ARS: Angular 5 ↔ Socket.IO ↔ Yii2 —
real-time + REST



CQRS-lite

separate read (reports) and write
(dialogue logs) paths



RBAC + token sessions

tariff-based access, token_dies, TTL
1800 in AliceBot



Core reuse

AliceBot and the HR bot share a
common core (composer:
webguru/hrbot)

Project map by industry and technology

At a glance: what / for whom / on what

AliceBot.pro	<i>SaaS · voice</i>	Phalcon · Multi-platform	HR survey bot	<i>HR · retail</i>	Phalcon · 4 channels
ARS hotline	<i>Security</i>	Yii2 · Angular · Socket.IO	vl-taxi.ru	<i>Taxi service</i>	Phalcon · SEO · CMS
karpala.ru	<i>Taxi marketplace</i>	Yii2 · Android	startbiz.space	<i>HR · freight</i>	Fat-Free · Telegram
BraidsBot	<i>E-commerce</i>	Symfony 4 · Doctrine	Centereko_Bot	<i>Medicine</i>	Symfony 4 · EasyAdmin
CRM Raskleika	<i>Field service</i>	F3 · AJAX · Java	Openpolice_ruBot	<i>Legal</i>	Native PHP · RSS

T E S T I M O N I A L S · 1 4

Customer testimonials

3 letters of appreciation

AO "MegaFon Retail" · OOO "Alliance Retail Security" · OOO CS "Garant"



Testimonial — "MegaFon Retail"

“

"AO "MegaFon Retail", represented by HR Director Alexander I. Bozhkov, expresses its gratitude to the WEBGURU PRO team (OOO Sparta) for the prompt and professional roll-out of a chatbot.

During the project the team resolved issues and delivered changes quickly. The product has fully met our expectations as a client. Within less than a month of active use, we are already seeing positive feedback from candidates.

Managers Anton Alekseev and Sergey Emelyanov addressed every question that came up during the work promptly and professionally."

HR Director · A. I. Bozhkov



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ИНН / КПП 7825695758 / 774901001

23.01.2018 № 12-НР-Учк-00001/18

на _____ от _____

Благодарственное письмо

Акционерное общество «Мегасон Ритейл» в лице Директора по персоналу Божкова Александра Игоревича выражает благодарность коллективу компании WEBGURU PRO (ООО СПАРТА) за оперативное и качественное внедрение чат-бота. В процессе реализации проекта коллеги быстро и оперативно решали вопросы и вносили корректировки. Сам продукт полностью удовлетворил нас, как заказчиков. За неполный месяц активного использования у нас уже есть позитивный отклик от соискателей. Менеджеры Антон Алексеев и Сергей Емельянов оперативно и профессионально решили все возникающие в процессе работы вопросы.

С Уважением
АО «Мегасон Ритейл»
Директор по персоналу Божков А.И.



Testimonial — "Alliance Retail Security"

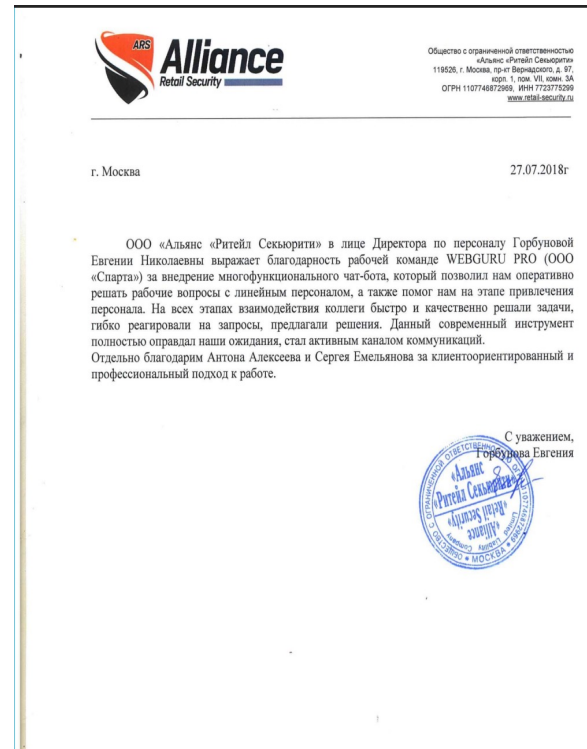
“

"ООО "Alliance Retail Security", represented by HR Director Evgeniya N. Gorbunova, expresses its gratitude to the WEBGURU PRO team (ООО "Sparta") for delivering a multi-purpose chatbot that lets us address day-to-day issues with front-line staff and helped us with recruitment as well.

At every stage of the project the team resolved tasks quickly and thoroughly, responded flexibly to requests and proposed solutions. This modern tool has fully met our expectations and has become an active communication channel.

Our separate thanks go to Anton Alekseev and Sergey Emelyanov for their client-focused and professional approach."

HR Director · E. N. Gorbunova



Testimonial — CS "Garant"

“

"ООО CS "Garant", represented by General Director Ruslan A. Tillaev, expresses its gratitude to the WEBGURU PRO team (ООО "Sparta") for productive cooperation, timeliness, responsiveness and responsibility while delivering our order.

We are very pleased with the competent and professional staff of your company, and in particular with account manager Alena Nametkina, who handled our project quickly and efficiently at all times."

General Director · R. A. Tillaev



ООО «ЦС «ГАРАНТ»

ИНН/КПП 1606006771/160501001
420021, Республика Татарстан, г. Казань, ул. Ахтиянова, д.1.
р/сч. 40702810600380517000
Филиал «Самара» КВ «ЛОКО-Банк» (ЗАО) г.Самара.
К/с (в валюте РФ) №301010110500000000925
БИК 043601925

г. Казань

ИСх №568/01.12.2017

От ООО «ЦС «ГАРАНТ»

Благодарственное письмо

Общество с ограниченной ответственностью «ЦС «ГАРАНТ», в лице генерального директора Тиллаева Руслана Абдулхамидовича, выражает благодарность коллективу компании WEBGURU PRO (ООО «Спарта») за плодотворное сотрудничество, своевременность, оперативность, ответственность при выполнении нашего заказа. Очень довольны компетентным и профессиональным персоналом Вашей компании, в частности менеджером Аленой Наметкиной, которая занималась нашим проектом, всегда быстро и оперативно.



С Уважением

ООО «ЦС «Гарант»

Ген. директор Тиллаев Р.А.

CTO responsibilities



Architecture design

framework choice, DB model, API structure, integration points



Core-module development

hands-on work on critical pieces: middleware, billing, the builder



Code review and standards

reviewing all team code, enforcing PSR, PHPUnit, Codeception, phpstan



Team mentoring

developer onboarding, pair programming, post-incident reviews



DevOps and infrastructure

deployment, server configuration, CI/CD setup, logging



Client communication

technical interviews with clients, scoping, estimates, result demos

Thank you

PROJECT STATUS

All projects are closed or on hold — the portfolio is historical. A public link remains for AliceBot.pro · spark.ru/startup/webguru-pro/blog/45356



Sergey B. Emelyanov

CTO at Webguru.pro (2017–2020) · Senior PHP / Go Developer



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portfolio · GitHub